



Assessment Task 3

Teacher

Section 1

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1 What is i^{2022} ?

- A. 1
- B. i
- C. -1
- D. $-i$

Questions 2 and 3 both refer to the following statement and selection of answers.

Statement

“If I go to *Efficiency Coaching* I will become efficient.”

Answers

- A. If I do not go to *Efficiency Coaching* I will not become efficient.
- B. If I did not become efficient I did not go to *Efficiency Coaching*.
- C. I went to *Efficiency Coaching* and I did not become efficient.
- D. I did not go to *Efficiency Coaching* and I became efficient.

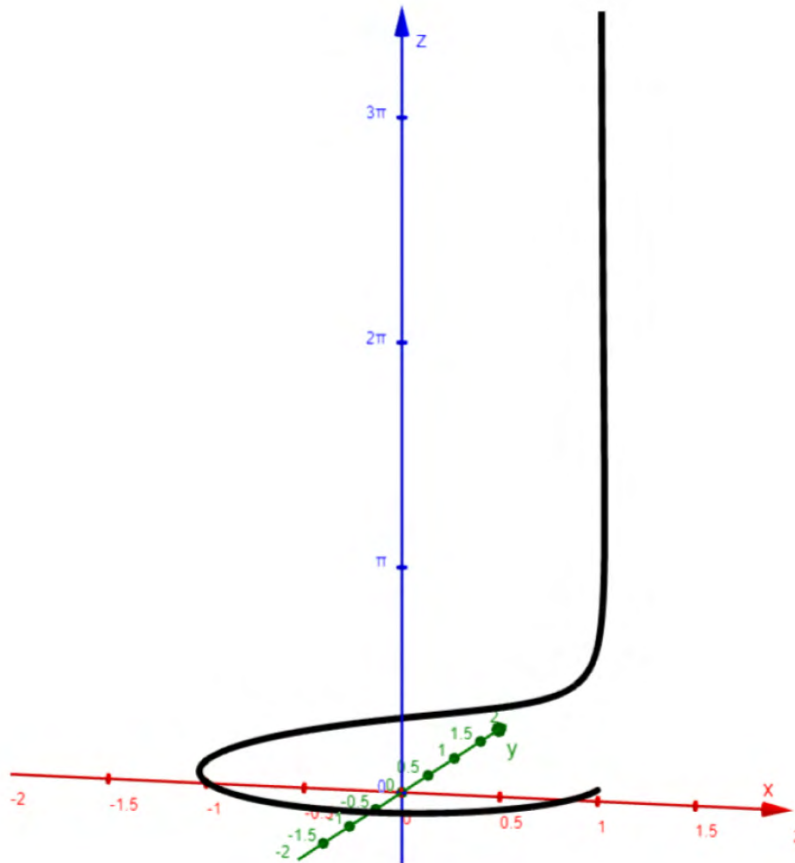
2 Which of the above is the negation of the statement?

3 Which of the above is the contrapositive of the statement?

4 Which of the following describes a set of points, $z \in \mathbb{C}$, that lie on a circle with a radius of 5 and a centre of (0,0)?

- A. $z\bar{z} = 5$
- B. $z^2 = 25$
- C. $\operatorname{Re}(z^2) + \operatorname{Im}(z^2) = 25$
- D. $(z + \bar{z})^2 - (z - \bar{z})^2 = 100$

5



Which of the following parametric equations best describes the graph above.

- A. $x = \cos t$, $y = \sin t$, $z = t$
- B. $x = \cos t$, $y = \sin t$, $z = \frac{1}{t}$
- C. $x = \cos t$, $y = t$, $z = \sin t$
- D. $x = \cos t$, $y = \frac{1}{t}$, $z = \sin t$

6 Evaluate:

$$\int_{-a}^a \frac{e^x}{e^x + e^{-x}} dx$$

- A. 0
- B. a
- C. $\tan^{-1}(e^a) - \tan^{-1}(e^{-a})$
- D. $\ln(e^{2a} - 1) - a$

- 7 The position vector of the centre of a sphere is $\underline{c} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ and the sphere is defined such that $|\underline{p} - \underline{c}| = 3$, where $\underline{p}$ gives the position vectors of the points on the surface of the sphere. Which of the following is the cartesian equation of a sphere which is tangent to $\underline{p}$?

- A. $(x - 5)^2 + (y - 7)^2 + (z + 2)^2 = 36$
- B. $(x - 3)^2 + (y - 3)^2 + (z - 2)^2 = 9$
- C. $(x - 4)^2 + (y - 2)^2 + (z - 8)^2 = 18$
- D. $x^2 + y^2 + z^2 = 21$

- 8 If z lies on the circle $|z| = 2$, find the minimum value of:

$$\left| \frac{1}{z^4 - 4z^2 + 3} \right|$$

- A. $\frac{1}{35}$
- B. $\frac{1}{19}$
- C. $\frac{1}{3}$
- D. $\frac{1}{2}$

- 9 A particle is initially stationary at position $x = 1$ metres. It experiences an acceleration of $\ddot{x} = 2x - 4$ m/s².

What is the speed of the particle after it has travelled for a total of 5 metres?

- A. $2\sqrt{2}$ m/s
- B. $\sqrt{30}$ m/s
- C. $\sqrt{70}$ m/s
- D. $4\sqrt{6}$ m/s

10 Which of the following is a possible solution to $\sin x = 2$?

A. $4i$

B. $\frac{i\sqrt{2}}{1+\sqrt{2}} - \pi$

C. $\pi - e^{i\sqrt{2}}$

D. $\frac{\pi}{2} + i \ln(2 + \sqrt{3})$

Section 2

90 Marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

For questions in Section 2, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)

(a) $z = 1 + 2i$ is a solution to the cubic equation $2z^3 - z^2 + 4z + k = 0$, where k is a real number.

(i) Find the other two solutions. 2

(ii) Evaluate k . 1

(b) Points A , B and C are given by the position vectors $\underline{a} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$ and $\underline{c} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$ respectively.
Find:

(i) $|\overrightarrow{AB}|$ and $|\overrightarrow{BC}|$ 2

(ii) $\angle ABC$ 2

(iii) The exact area of $\triangle ABC$ 2

(c) Shade the region on the Argand diagram that is satisfied by both the inequalities

$|z - 3 + i| \leq 3$ and $|z| \geq |z - 2i|$ 3

(d) Prove that $\sqrt{3k + 2}$ is not an integer for all positive integers k . 3

Question 12 (15 Marks)

- (a) The displacement of a particle is given by $x = \sqrt{3} \sin 2t - \cos 2t + 3$.
- (i) Find the initial displacement and velocity of the particle. **2**
- (ii) Prove that the particle is moving in simple harmonic motion. **1**
- (iii) Find v^2 in terms of x . **2**
- (iv) Hence or otherwise find the maximum displacement and maximum speed of the particle. **2**
- (b) Find the point on the line $\underline{r} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ that is closest to the origin. **3**
- (c) On a standard Argand diagram, the complex number $\sqrt{3} + i$ represents one of the vertices of a regular hexagon, with centre at the origin O .
- (i) Find the complex numbers which represent the other 5 vertices **2**
- (ii) The complex numbers that represent the vertices are all raised to the power of 4, creating a closed shape S , whose sides are straight line segments.
- Find the area of S . **3**

Question 13 (15 Marks)

- (a) By considering the partial fraction

$$\frac{1}{x^4 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{C}{x^2 + 1}$$

find:

$$\int \frac{1}{x^4 - 1} dx$$

3

- (b) Find

$$\int x^2 \sin x dx$$

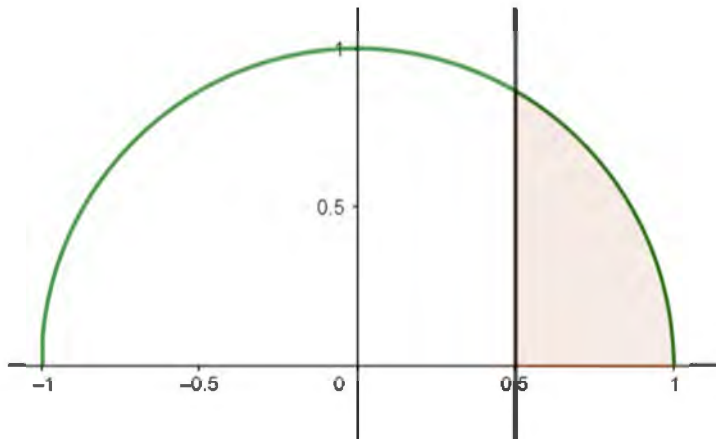
3

- (c) Use an appropriate substitution to evaluate:

$$\int_0^{\frac{\pi}{2}} \frac{4}{3 + 5\cos x} dx$$

3

- (d) The diagram below shows $y = \sqrt{1 - x^2}$ and $x = \frac{1}{2}$



Find the exact shaded area enclosed between $y = \sqrt{1 - x^2}$, $x = \frac{1}{2}$ and the x -axis.

3

- (e) Given that $a = 3^{-\frac{1}{6}}$ and $b = 3^{\frac{1}{6}}$, using the substitution $u = \frac{1}{x}$ or otherwise, evaluate:

$$\int_a^b \frac{x^3}{(1 + x)(1 + x^6)} dx$$

3

Question 14 (15 Marks)

(a) (i) $f(z) = z^6 + 8z^3 + 64$, $z \in \mathbb{C}$

Given that $f(z) = 0$, show that

$$z^3 = -4 \pm 4\sqrt{3}i \quad 1$$

(ii) Hence or otherwise find all six solutions to the equation $f(z) = 0$, giving answers in the form $z = re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$. 2

(iii) Hence show that:

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{6\pi}{9} + \cos \frac{8\pi}{9} = -\frac{1}{2} \quad 3$$

- (b) Mr Berry confiscates an *Efficiency Coaching* booklet from a student (who should not have been doing their tutoring homework in class) and drops it from the window of B14.

The booklet lands on the playground exactly 1.5 seconds later.

The booklet weighs 0.2 kg and experiences a force due to air resistance of magnitude $0.4v$ Newtons, where v is its velocity.

Letting gravity be 9.8 m/s^2 , find to two decimal places:

(i) The speed of the booklet when it hits the ground 3

(ii) The percentage of its terminal velocity it reached at the moment of impact 1

(iii) The height the booklet was dropped from 3

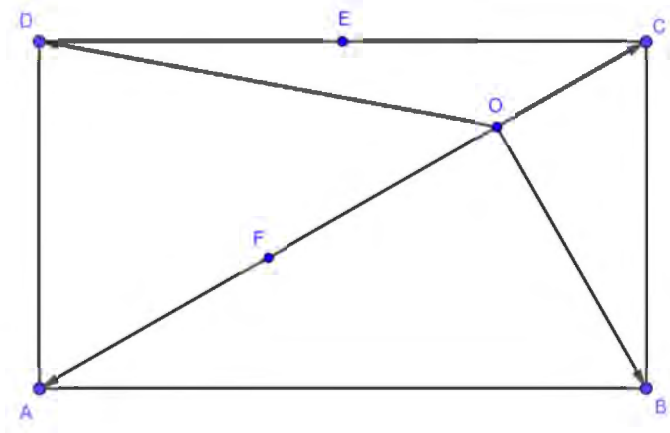
(c) Given that $z \in \mathbb{C}$ such that $\text{Im}(z) \neq 0$ and $\frac{z}{1+z^2}$ is a real number. Show that $|z| = 1$ 2

Question 15 (15 Marks)

- (a) In the diagram below, O is the origin and points A, B, C and D have position vectors $\underline{a}$, $\underline{b}$, $\underline{c}$ and $\underline{d}$ respectively.

E is the midpoint of CD and F is the midpoint of OA .

$ABCD$ is a rectangle such that O lies on diagonal AC , and OB is perpendicular to AC



- (i) Show that $|\underline{b}|^2 + \underline{a} \cdot \underline{c} = 0$ 2
- (ii) Hence, or otherwise, show that $\angle BFE = \frac{\pi}{2}$ 3
- (b) (i) Expand and simplify $3[(x - 2)^3 + 2]$ 1

(ii) Given

$$I_n = \int_0^a \frac{x^n}{\sqrt{a^2 - x^2}} dx, \quad n \in \mathbb{N}$$

clearly show that:

$$I_n = \frac{a^2(n-1)}{n} I_{n-2}, \quad n \geq 2$$
 3

(iii) Show that

$$\int_2^4 \frac{3x^3 - 18x^2 + 36x - 18}{\sqrt{4x - x^2}} dx = 3I_3 + 3\pi$$
 4

(iv) Hence or otherwise, evaluate

$$\int_2^4 \frac{3x^3 - 18x^2 + 36x - 18}{\sqrt{4x - x^2}} dx$$
 2

Question 16 (15 Marks)

- (a) (i) By considering the expansion of $k^3 - (k - 1)^3$ show that:

$$\sum_{k=1}^n k^2 = \frac{n}{6}(n+1)(2n+1) \quad 2$$

(do **NOT** use mathematical induction)

- (ii) Prove that there are no finite sets of prime numbers where $2^2 + 3^2 + \dots + n^2$ is a multiple of at least one element of the set for all integers $n \geq 2$. 2

- (b) (i) Let $\underline{w} = \underline{u} + r\underline{v}$ where $\underline{w}$, $\underline{u}$ and $\underline{v}$ are vectors and r is a scalar.

Prove that $|\underline{v}|^2 r^2 + 2(\underline{u} \cdot \underline{v})r + |\underline{u}|^2 \geq 0$ 1

- (ii) Use your result from (i) to show that $\underline{u} \cdot \underline{v} \leq |\underline{u}||\underline{v}|$ 2

- (iii) Use your result from part (ii) to show that for real numbers x, y and z .

$$xy + yz + zx \leq x^2 + y^2 + z^2 \quad 1$$

- (iv) Use your result from part (iii) to show that for non-negative values of x, y and z :

$$xyz \leq \frac{x^3 + y^3 + z^3}{3} \quad 1$$

- (v) Given that $a + b + c + d = 0$ and $a^2 + b^2 + c^2 + d^2 = 12$. Find:

1. The maximum value of $abcd$ 2

2. The minimum value of $abcd$ 4

1 What is i^{2022} ?

A. 1

B. i

C. -1

D. $-i$

$$\begin{aligned} i^{2022} &= i^{2020} \times i^2 \\ &= ((i^4)^{505}) \times -1 \\ &= 1^{505} \times -1 \\ &= 1 \times -1 \\ &= -1 \end{aligned}$$

Questions 2 and 3 both refer to the following statement and selection of answers.

Statement

"If I go to *Efficiency Coaching* I will become efficient."

Answers

- A. If I do not go to *Efficiency Coaching* I will not become efficient.
- B. If I did not become efficient I did not go to *Efficiency Coaching*.
- C. I went to *Efficiency Coaching* and I did not become efficient.
- D. I did not go to *Efficiency Coaching* and I became efficient.

2 Which of the above is the negation of the statement?

C

3 Which of the above is the contrapositive of the statement?

B

4 Which of the following describes a set of points, $z \in \mathbb{C}$, that lie on a circle with a radius of 5 and a centre of (0,0)?

A. $z\bar{z} = 5$

$r = \sqrt{5}$

B. $z^2 = 25$

not a circle

C. $\operatorname{Re}(z^2) + \operatorname{Im}(z^2) = 25$

not a circle

D. $(z + \bar{z})^2 - (z - \bar{z})^2 = 100$

$$z = x + iy$$

$$z + \bar{z} = 2x$$

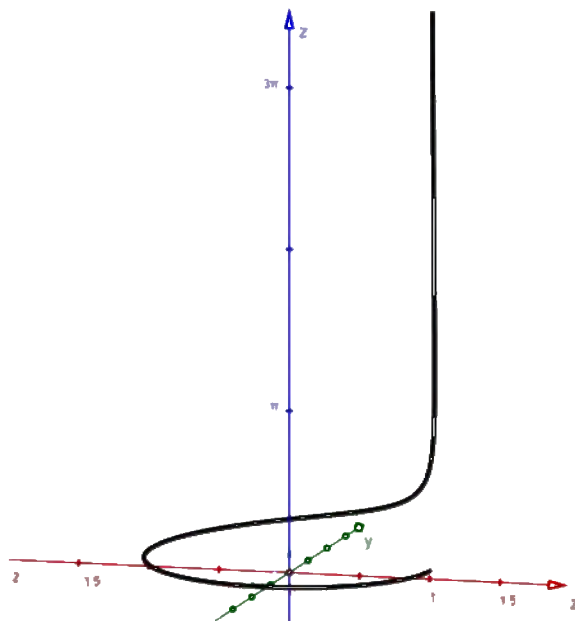
$$z - \bar{z} = 2iy$$

$$(2x)^2 - (2iy)^2$$

$$4x^2 + 4y^2 = 100$$

$$x^2 + y^2 = 25$$

5



at $t = 0$ we see
 $x = 1$, $y = 0$ &
 z is undefined

Which of the following parametric equations best describes the graph above.

A. $x = \cos t$, $y = \sin t$, $z = t$

B. $x = \cos t$, $y = \sin t$, $z = \frac{1}{t}$

C. $x = \cos t$, $y = t$, $z = \sin t$

D. $x = \cos t$, $y = \frac{1}{t}$, $z = \sin t$

6

Evaluate:

$$\int_{-a}^a \frac{e^x}{e^x + e^{-x}} dx$$

$$\int_{-a}^a \frac{e^x}{e^x + e^{-x}} dx$$

A. 0

B. a

C. $\tan^{-1}(e^a) - \tan^{-1}(e^{-a})$

D. $\ln(e^{2a} - 1) - a$

$$= \int_0^a \frac{e^x}{e^x + e^{-x}} dx + \int_0^a \frac{e^{-x}}{e^{-x} + e^x} dx$$

$$= \int_0^a \frac{e^x + e^{-x}}{e^x + e^{-x}} dx$$

$$= \int_0^a 1 dx$$

$$= [x]_0^a$$

$$= a$$

- 7 The position vector of the centre of a sphere is $\zeta = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ and the sphere is defined such that $|\underline{v} - \zeta| = 3$, where $\underline{v}$ gives the position vectors of the points on the surface of the sphere. Which of the following is the cartesian equation of a sphere which is tangent to $\underline{v}$?

radius

A. $(x-5)^2 + (y-7)^2 + (z+2)^2 = 36$ ← radius = 6

B. $(x-3)^2 + (y-3)^2 + (z-2)^2 = 9$

C. $(x-4)^2 + (y-2)^2 + (z-8)^2 = 18$

D. $x^2 + y^2 + z^2 = 21$

sum of radii = $6 + 3 = 9$

distance of centres

$$d^2 = (2-5)^2 + (1-7)^2 + (4-(-2))^2$$

$$= 3^2 + 6^2 + 6^2$$

$$= 81$$

$d = 9$

distance between centres = sum of radii

- 8 If z lies on the circle $|z| = 2$, find the minimum value of:

$$\left| \frac{1}{z^4 - 4z^2 + 3} \right|$$

A. $\frac{1}{35}$

B. $\frac{1}{19}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

$$\frac{1}{|z^4 - 4z^2 + 3|}$$

$$= \frac{1}{|z^2 - 3| |z^2 - 1|}$$

$$\geq \frac{1}{(|z^2| + |-3|)(|z^2| + |-1|)}$$

triangle inequality

$$= \frac{1}{(4+3)(4+1)}$$

$$= \frac{1}{7 \times 5}$$

$$= \frac{1}{35}$$

A particle is initially stationary at position $x = 1$ metres. It experiences an acceleration of $\ddot{x} = 2x - 4 \text{ m/s}^2$.

What is the speed of the particle after it has travelled for a total of 5 metres?

A. $2\sqrt{2} \text{ m/s}$

B. $\sqrt{30} \text{ m/s}$

☒ C. $\sqrt{70} \text{ m/s}$

D. $4\sqrt{6} \text{ m/s}$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x - 4$$

$$\frac{d}{dx} v^2 = 4x - 8$$

$$[v^2]_0^5 = [2x^2 - 8x]_1^5$$

$$v^2 = 2x^2 - 8x - (2 - 8)$$

$$v^2 = 2x^2 - 8x + 6$$

$$= 2(x^2 - 4x + 3)$$

$$= 2(x-1)(x-3)$$

$$\therefore v = \pm \sqrt{2(x-1)(x-3)}$$

$$v = \pm \sqrt{2 \times (-4-1)(-4-3)}$$

$$= \pm \sqrt{70}$$

$$\therefore \text{speed} = \sqrt{70} \text{ m/s}$$

as initial velocity is 0 and acceleration is negative particle always moves left so $x = -4$ when it has moved 5 metres

10 Which of the following is a possible solution to $\sin x = 2$?

A. $4i$

B. $\frac{i\sqrt{2}}{1+\sqrt{2}} - \pi$

C. $\pi - e^{i\sqrt{2}}$

D. $\frac{\pi}{2} + i \ln(2 + \sqrt{3})$

$$\frac{e^{ix} - e^{-ix}}{2i} = 2$$

$$e^{ix} - \frac{1}{e^{ix}} = 4i$$

$$(e^{ix})^2 - 1 = (4i)e^{ix}$$

$$(e^{ix})^2 - 4i(e^{ix}) - 1 = 0$$

$$e^{ix} = \frac{4i \pm \sqrt{(4i)^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{4i \pm \sqrt{-16 + 4}}{2}$$

$$= \frac{4i \pm \sqrt{-12}}{2}$$

$$= \frac{4i \pm 2\sqrt{3}i}{2}$$

$$e^{ix} = (2 \pm \sqrt{3})i$$

$$ix = \log_e[(2 \pm \sqrt{3})i]$$

$$= \log_e i + \log_e(2 \pm \sqrt{3})$$

$$= \log_e e^{i\frac{\pi}{2}} + \log_e(2 \pm \sqrt{3})$$

$$ix = i\left(\frac{\pi}{2}\right) + \log_e(2 \pm \sqrt{3})$$

$$x = \frac{\pi}{2} + \frac{\log_e(2 \pm \sqrt{3})}{i} \times \frac{i}{i}$$

$$= \frac{\pi}{2} - i \log_e(2 \pm \sqrt{3})$$

$$= \frac{\pi}{2} - i \log_e(2 - \sqrt{3})$$

one of the solutions

$$= \frac{\pi}{2} - i \log_e\left(\frac{1}{2 + \sqrt{3}}\right)$$

$$= \frac{\pi}{2} - i \log_e(2 + \sqrt{3})^{-1}$$

$$= \frac{\pi}{2} + i \log_e(2 + \sqrt{3})$$

Question 11 (15 Marks)

(a) $z = 1 + 2i$ is a solution to the cubic equation $2z^3 - z^2 + 4z + k = 0$, where k is a real number.

(i) Find the other two solutions.

2

$$\alpha = 1 + 2i$$

$$\beta = 1 - 2i \text{ (conjugate root theorem)}$$



$$\alpha + \beta + \gamma = -\left(-\frac{1}{2}\right)$$

$$\therefore 1 + \cancel{2i} + 1 - \cancel{2i} + \gamma = \frac{1}{2}$$

$$2 + \gamma = \frac{1}{2}$$

$$\gamma = -\frac{3}{2}$$



(ii) Evaluate k .

1

$$\alpha \beta \gamma = -\frac{k}{2}$$

$$(1 + 2i)(1 - 2i)\left(\cancel{-\frac{3}{2}}\right) = \cancel{-\frac{k}{2}}$$

$$\therefore k = 3(1 + 2i)(1 - 2i)$$

$$= 3(1^2 + 2^2)$$

$$= 3 \times 5$$

$$= 15$$



- (b) Points A , B and C are given by the position vectors $\vec{a} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$ and $\vec{c} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$ respectively.

Find:

- (i) $|\vec{AB}|$ and $|\vec{BC}|$

2

$$\begin{aligned} |\vec{AB}|^2 &= (1-2)^2 + (4-5)^2 + (-2-1)^2 \\ &= (-1)^2 + (-1)^2 + (-3)^2 \\ &= 11 \\ \therefore |\vec{AB}| &= \sqrt{11} \end{aligned}$$



$$\begin{aligned} |\vec{BC}|^2 &= (2-(-1))^2 + (5-1)^2 + (1-4)^2 \\ &= 3^2 + 4^2 + 3^2 \\ &= 34 \\ \therefore |\vec{BC}| &= \sqrt{34} \end{aligned}$$



- (ii) $\angle ABC$

2

$$\vec{BA} \cdot \vec{BC} = |\vec{BA}| |\vec{BC}| \cos \theta$$

$$\therefore \cos \theta = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{AB}| |\vec{BC}|}$$



$$= \frac{(-1)(-3) + (-1)(-4) + (-3)(3)}{\sqrt{11} \times \sqrt{34}}$$

$$= \frac{-2}{\sqrt{374}}$$

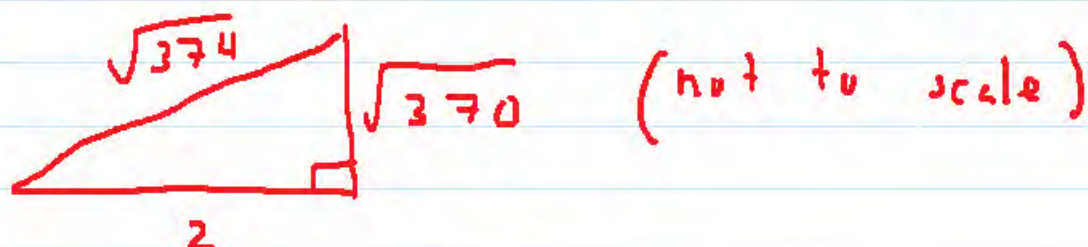
$$\vec{BA} = \begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix}$$

$$\vec{BC} = \begin{pmatrix} -3 \\ -4 \\ 3 \end{pmatrix}$$

$$\therefore \angle ABC = \cos^{-1} \left(\frac{-2}{\sqrt{374}} \right)$$



$$\begin{aligned}\text{Area} &= \frac{1}{2} |\vec{AB}| |\vec{BC}| \sin \angle ABC \\ &= \frac{1}{2} \times \sqrt{11} \times \sqrt{34} \times \sin \left[\cos^{-1} \left(\frac{2}{\sqrt{374}} \right) \right] \quad \checkmark\end{aligned}$$



$$= \frac{1}{2} \times \sqrt{\cancel{374}} \times \frac{\sqrt{370}}{\sqrt{\cancel{374}}}$$

$$= \frac{\sqrt{370}}{2} \text{ units}^2 \quad \checkmark$$

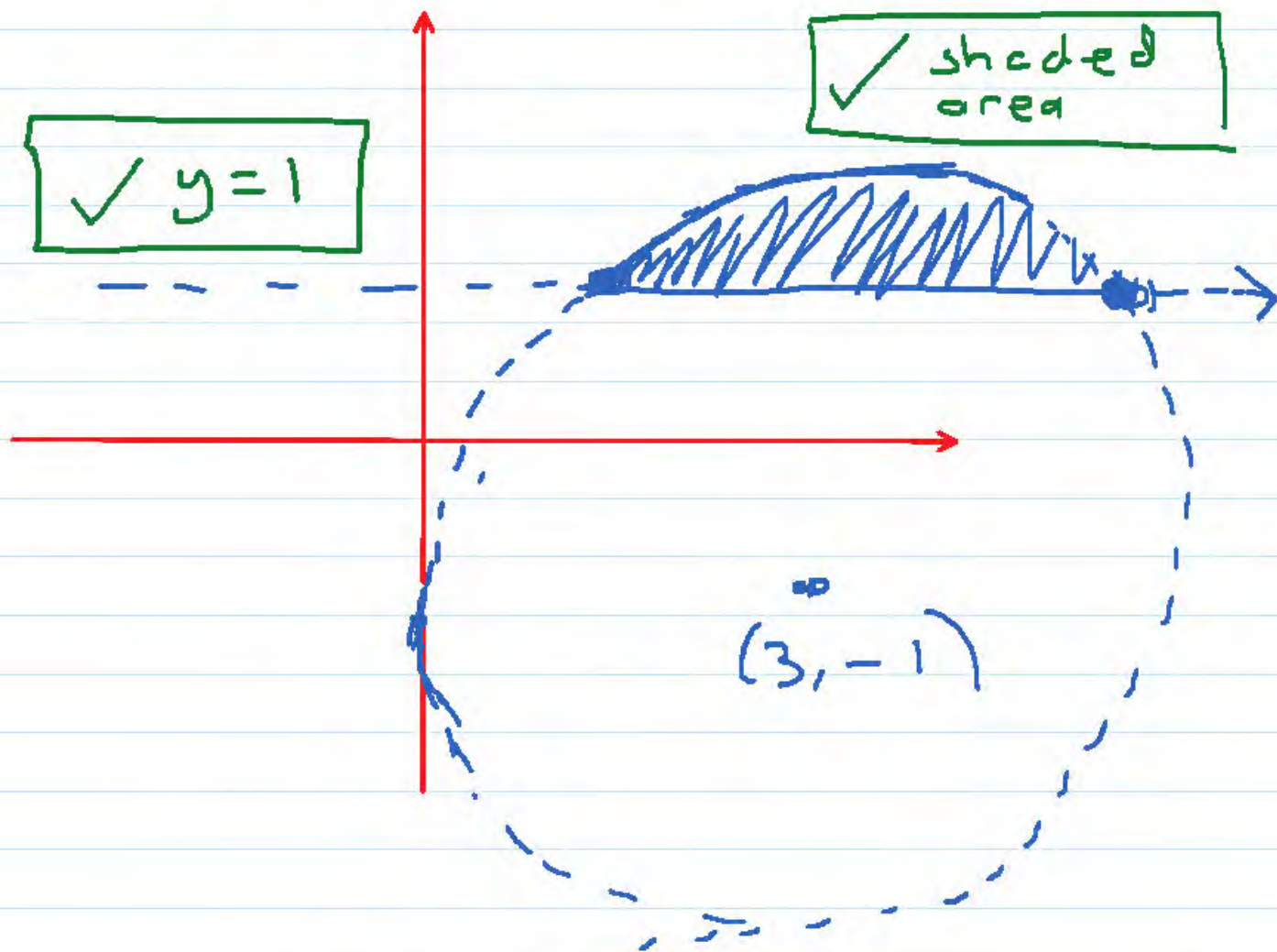
(c) Shade the region on the Argand diagram that is satisfied by both the inequalities

$$|z - 3 + i| \leq 3 \text{ and } |z| \geq |z - 2i|$$

3

$|z - (3 - i)| \leq 3$ is inside a circle with centre $(3, -1)$ & radius 3

$|z| \geq |z - 2i|$ is above the perpendicular bisector of $y = 0$ & $y = 2$.



$$\checkmark (x-3)^2 + (y+1)^2 = 3$$

Assume integer P exists such that

$$p = \sqrt{3k+2}$$

$$\therefore p^2 = 3k+2$$

✓ assumption

case 1 $p = 3n$ ($n \in \mathbb{Z}$)

$$\therefore (3n)^2 = 3k+2$$

$$9n^2 = 3k+2$$

$$3(3n^2) = 3k+2 \quad \left(\begin{array}{l} \text{impossible as LHS is a} \\ \text{multiple of 3 while RHS is not} \end{array} \right)$$

case 2 $p = 3n+1$ ($n \in \mathbb{Z}$)

✓ at least 1 case

$$(3n+1)^2 = 3k+2$$

$$9n^2 + 6n + 1 = 3k+2$$

$$3(3n^2 + 2n) + 1 = 3k+2$$

(impossible as LHS is one more than a multiple of 3 while RHS is 2 more)

case 3 $p = 3n+2$ ($n \in \mathbb{Z}$)

$$(3n+2)^2 = 3k+2$$

$$9n^2 + 12n + 4 = 3k+2$$

$$3(3n^2 + 4n + 1) + 1 = 3k+2$$

(impossible since LHS is one more than a multiple of 3 while RHS is 2 more)

$\therefore$ there are no possible values for p where $p^2 = 3k+2$

proof by contradiction

✓ all cases and conclusion

Question 12 (15 Marks)

(a) The displacement of a particle is given by $x = \sqrt{3} \sin 2t - \cos 2t + 3$.

(i) Find the initial displacement and velocity of the particle.

2

when $t = 0$

$$\begin{aligned} x &= \sqrt{3} \sin(2(0)) - \cos(2(0)) + 3 \\ &= 0 - 1 + 3 \\ &= 2 \text{ m} \end{aligned}$$



$$\dot{x} = 2\sqrt{3} \cos 2t + 2 \sin 2t$$

when $t = 0$

$$\begin{aligned} \dot{x} &= 2\sqrt{3} \cos(2(0)) + 2 \sin(2(0)) \\ &= 2\sqrt{3} \text{ m/s} \end{aligned}$$



(ii) Prove that the particle is moving in simple harmonic motion.

1

$$\begin{aligned} \ddot{x} &= -4\sqrt{3} \sin 2t + 4 \cos 2t \\ &= -4(\sqrt{3} \sin 2t - \cos 2t) \\ &= -4(x - 3) \end{aligned}$$



$\therefore$ particle is moving in simple harmonic motion as acceleration is proportional to displacement in the opposite direction

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -4(x-3)$$

$$\frac{d}{dx} v^2 = -8(x-3)$$

$$[v^2]_{2\sqrt{3}}^x = \int_{2\sqrt{3}}^x -8x + 24 \, dx$$

$$v^2 - (2\sqrt{3})^2 = [-4x^2 + 24x]_{2\sqrt{3}}^x$$

$$\begin{aligned} v^2 - 12 &= -4x^2 + 24x - (-4 \times 2^2 + 24(2)) \\ &= -4x^2 + 24x - 32 \end{aligned}$$

$$\therefore v^2 = -4x^2 + 24x - 20$$

✓ integration

✓ using initial conditions to get answer

(iv) Hence or otherwise find the maximum displacement and maximum speed of the particle.

2

$$v^2 = -4(x^2 - 6x + 5)$$

$$= -4(x-5)(x-1)$$

$$\therefore v^2 = 0 \text{ at } x = 5 \text{ and } x = 1$$

$$\therefore \text{max displacement at } x = 5$$

$$v^2 \text{ is maximised at } \frac{5+1}{2} = 3$$

$$\begin{aligned} \therefore v^2 &= -4(3-5)(3-1) \\ &= 16 \end{aligned}$$

$$\therefore \text{max speed} = 4 \text{ m/s}$$

✓

✓

(b) Find the point on the line $\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ that is closest to the origin.

3

$$d^2 = (1+2\lambda)^2 + (4+\lambda)^2 + (6+\lambda)^2$$

$$= 1 + 4\lambda + 4\lambda^2 + 16 + 8\lambda + \lambda^2 + 36 + 12\lambda + \lambda^2$$

$$= 6\lambda^2 + 24\lambda + 53$$

find λ to minimise d^2 (and hence d)

$$-\frac{24}{2(6)} = -\frac{24}{12}$$

$$= -2$$

$$\therefore \text{ point is } \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix}$$

✓ distance
formula

✓ minimise
distance

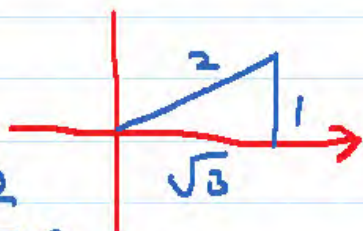
✓ answer

- (c) On a standard Argand diagram, the complex number $\sqrt{3} + i$ represents one of the vertices of a regular hexagon, with centre at the origin O .

(i) Find the complex numbers which represent the other 5 vertices

2

$$\sqrt{3} + i$$



$$\text{modulus} = 2$$

$$\text{argument } \pi/6$$

$$\sqrt{3} + i = 2e^{i\pi/6}$$

✓ euler form

✓ vertices

∴ vertices are

$$2e^{i\pi/6}, 2e^{5\pi/6}, 2e^{-\pi/6}, 2e^{-5\pi/6}, 2e^{i\pi/2}$$

(found by adding $\frac{2\pi}{6} = \frac{\pi}{3}$ to the argument)

- (ii) The complex numbers that represent the vertices are all raised to the power of 4, creating a closed shape S , whose sides are straight line segments.

Find the area of S .

3 ✓ value

Raising answers in (i) to 4 we get

$$16e^{2\pi i}, 16e^{10\pi i}, 16e^{-2\pi i}, 16e^{-10\pi i}, 16e^{2\pi i}$$

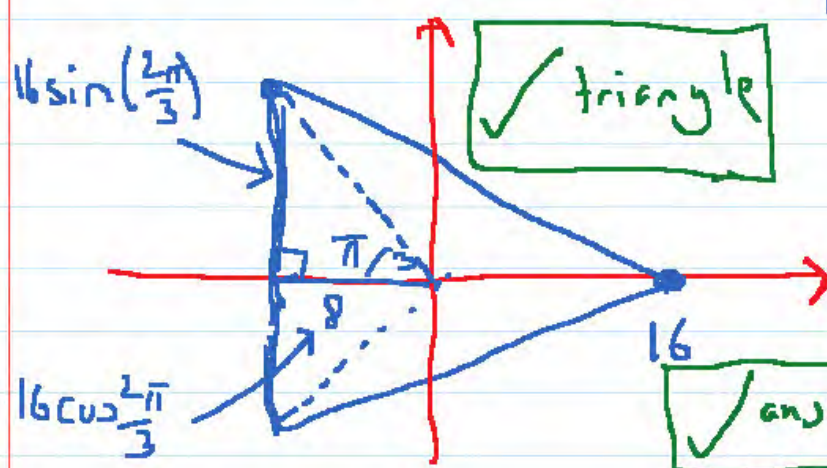
noting the duplicates, 3 unique values create a triangle

$$\text{height} = 2 \times 16 \sin\left(\frac{\pi}{3}\right) = 16\sqrt{3}$$

$$\text{base} = 16 + 16 \times \cos\frac{\pi}{3} = 24$$

$$A = \frac{1}{2} \times 16\sqrt{3} \times 24$$

$$\text{✓ answer} = 192\sqrt{3} \text{ units}^2$$



✓ triangle

Question 13 (15 Marks)

(a) By considering the partial fraction

$$\frac{1}{x^4 - 1} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x^2+1}$$

find:

$$\int \frac{1}{x^4 - 1} dx$$

3

$$A(x+1)(x^2+1) + B(x-1)(x^2+1) + C(x^2-1) = 1$$

$$\underline{x = 1}$$

$$A(2)(2) + 0 + 0 = 1$$

$$A = 1/4$$

$$\underline{x = -1}$$

$$0 + B(-2)(2) = 1$$

$$B = -1/4$$

✓ A, B & C

$$\underline{x = 0}$$

$$\frac{1}{4}(1)(1) - \frac{1}{4}(-1)(1) + C(-1) = 1$$

$$\frac{1}{2} - C = 1$$

$$C = -\frac{1}{2}$$

✓ at least one correct integration

$$\int \frac{1}{x^4 - 1} dx = \frac{1}{4} \int \frac{1}{x-1} dx - \frac{1}{4} \int \frac{1}{x+1} dx - \frac{1}{2} \int \frac{1}{x^2+1} dx$$

$$= \frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| - \frac{1}{2} \tan^{-1}(x) + C$$

$$= \frac{1}{4} \ln \left| \frac{x-1}{x+1} \right| - \frac{1}{2} \tan^{-1}(x) + C$$

✓ answer

(b) Find

$$\int x^2 \sin x \, dx$$

$$\int \underbrace{x^2}_u \underbrace{\sin x}_{v'} \, dx = (-\underbrace{\cos x}_v) \underbrace{x^2}_u - \int \underbrace{2x}_{u'} (-\underbrace{\cos x}_v) \, dx$$

3
✓
1st
part

$$= -x^2 \cos x + 2 \int \underbrace{x}_u \underbrace{\cos x}_{v'} \, dx$$

$$= -x^2 \cos x + 2 \left[\underbrace{x}_u \underbrace{\sin x}_{v'} + \int \underbrace{\sin x}_{u'} \underbrace{dx}_v \right]$$

✓
2nd
part

$$= -x^2 \cos x + 2x \sin x - 2 \cos x + C$$

✓
Answer

(c) Use an appropriate substitution to evaluate:

$$\int_0^{\frac{\pi}{2}} \frac{4}{3+5\cos x} dx$$

sub into
integral

3

$$\int_0^{\frac{\pi}{2}} \frac{4}{3+5\cos x} dx$$

$$= \int_0^1 \frac{4}{3+5\left(\frac{1-t^2}{1+t^2}\right)} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{8}{3+3t^2+5-5t^2} dt$$

$$= \int_0^1 \frac{8}{8-2t^2} dt$$

$$= \int_0^1 \frac{\cancel{2} \times 4}{\cancel{2}(4-t^2)} dt$$

$$= \int_0^1 \frac{4}{4-t^2} dt$$

$$= \int_0^1 \left(\frac{1}{2+t} + \frac{1}{2-t} \right) dt$$

$$= \left[\ln|2+t| - \ln|2-t| \right]_0^1$$

$$= \left[\ln \left| \frac{2+t}{2-t} \right| \right]_0^1$$

$$= \ln 3 - \ln \left(\frac{2}{2} \right)$$

$$\text{let } t = \tan \frac{x}{2}$$

$$\therefore \frac{x}{2} = \tan^{-1} t$$

$$x = 2 + \tan^{-1} t$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$

$$\text{when } x = \frac{\pi}{2}$$

$$t = \tan \frac{\pi}{4} = 1$$

$$\text{when } x = 0$$

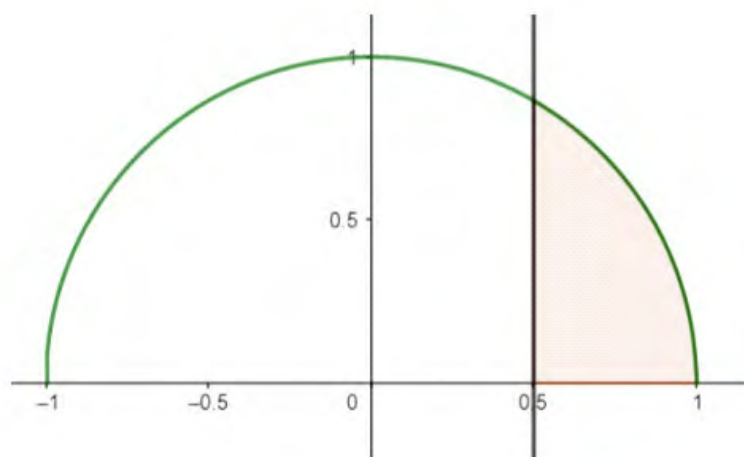
$$t = \tan 0 = 0$$

sub
limits

$$= \ln 3$$

✓ answer

- (d) The diagram below shows $y = \sqrt{1-x^2}$ and $x = \frac{1}{2}$



Find the exact shaded area enclosed between $y = \sqrt{1-x^2}$, $x = \frac{1}{2}$ and the x-axis.

3

$$A = \int_{0.5}^1 \sqrt{1-x^2} dx$$

$$= \int_{\pi/6}^{\pi/2} \underbrace{\sqrt{1-\sin^2 \theta}}_{\cos \theta} \cos \theta d\theta$$

$$= \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta$$

$$= \int_{\pi/6}^{\pi/2} \left(\frac{1}{2} + \frac{1}{2} \cos 2\theta \right) d\theta$$

$$= \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta \right]_{\pi/6}^{\pi/2}$$

$$= \left(\frac{\pi/2}{2} + \frac{1}{4} \sin \left(2 \times \frac{\pi}{2} \right) \right) - \left(\frac{\pi/6}{2} + \frac{1}{4} \sin \left(2 \times \frac{\pi}{6} \right) \right)$$

$$= \pi/4 - \pi/12 - 1/4 (\sqrt{3}/2)$$

$$x = \sin \theta$$

$$\frac{dx}{d\theta} = \cos \theta$$

$$dx = \cos \theta d\theta$$

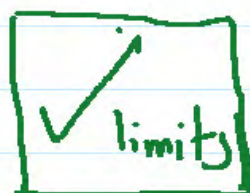
$$\theta = \sin^{-1} x$$

$$\text{when } x = 1$$

$$\theta = \frac{\pi}{2}$$

$$\text{when } x = 0.5$$

$$\theta = \frac{\pi}{6}$$



$$= \left(\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) \text{ units}^2$$

✓ answer

(e) Given that $a = 3^{-\frac{1}{6}}$ and $b = 3^{\frac{1}{6}}$, using the substitution $u = \frac{1}{x}$ or otherwise, evaluate:

$$\int_a^b \frac{x^3}{(1+x)(1+x^6)} dx$$

3

$$\begin{aligned} & \int_{3^{-1/6}}^{3^{1/6}} \frac{x^3}{(1+x)(1+x^6)} dx \\ &= \int_{3^{1/6}}^{3^{-1/6}} \frac{(1/u)^3}{(1+1/u)(1+1/u^6)} \left(-\frac{du}{u^2} \right) \quad \text{swap limits} \\ &= \int_{3^{-1/6}}^{3^{1/6}} \frac{1}{u^5(1+1/u)(1+1/u^6)} du \quad \text{substitution} \\ &= \int_{3^{-1/6}}^{3^{1/6}} \frac{1}{u^5(u+1)(u^4+1/u^2)} du \\ &= \int_{3^{-1/6}}^{3^{1/6}} \frac{1}{u^5(u+1)(u^4+u^{-2})} du \quad \times \frac{u^2}{u^2} \\ &= \int_{3^{-1/6}}^{3^{1/6}} \frac{u^2}{(1+u)(1+u^6)} du \quad u=x \\ &\therefore \int_{3^{-1/6}}^{3^{1/6}} \frac{x^3}{(1+x)(1+x^6)} dx = \int_{3^{-1/6}}^{3^{1/6}} \frac{x^2}{(1+x)(1+x^6)} dx \\ &\therefore 2 \int_{3^{-1/6}}^{3^{1/6}} \frac{x^3}{(1+x)(1+x^6)} dx = \int_{3^{-1/6}}^{3^{1/6}} \frac{x^3+x^2}{(1+x)(1+x^6)} dx \\ &\quad \checkmark \text{ addition of integrals} \end{aligned}$$

$u = \frac{1}{x}$
 $\frac{du}{dx} = -\frac{1}{x^2}$
 $dx = -du x^2$
 $= -\frac{du}{u^2}$
 when $x = 3^{1/6}$
 $u = 3^{-1/6}$
 when $x = 3^{-1/6}$
 $u = 3^{1/6}$

$$\int_{3^{-1/6}}^{3^{1/6}} \frac{x^3}{(1+x)(1+x^6)} dx = \frac{1}{2} \int_{3^{-1/6}}^{3^{1/6}} \frac{x^2 \cancel{(1+x)}}{\cancel{(1+x)}(1+x^6)} dx$$

$$= \frac{1}{2} \int_{3^{-1/6}}^{3^{1/6}} \frac{x^2}{1 + (x^3)^2} dx$$

$$= \frac{1}{6} \int_{3^{-1/6}}^{3^{1/6}} \frac{3x^2}{1 + (x^3)^2} dx$$

$$= \frac{1}{6} \left[\tan^{-1}(x^3) \right]_{3^{-1/6}}^{3^{1/6}}$$

$$= \frac{1}{6} \left[\tan^{-1} \sqrt{3} - \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \right]$$

$$= \frac{1}{6} \left[\frac{\pi}{3} - \frac{\pi}{6} \right]$$

↖ $3^{1/2}$
↖ $3^{-1/2}$

$$= \frac{1}{6} \times \frac{\pi}{6}$$

$$= \frac{\pi}{36}$$

✓ answer

(a) (i) $f(z) = z^6 + 8z^3 + 64 = 0$, $z \in \mathbb{C}$

Given that $f(z) = 0$, show that

$$z^3 = -4 \pm 4\sqrt{3}i$$

1

$$1(z^3)^2 + 8(z^3) + 64 = 0$$

$$z^3 = \frac{-8 \pm \sqrt{8^2 - 4(1)(64)}}{2(1)}$$

✓ quadratic formula

$$= \frac{-8 \pm \sqrt{162}}{2}$$

$$= \frac{-8 \pm 8\sqrt{3}}{2}$$

$$= -4 \pm 4\sqrt{3}i$$

(ii) Hence or otherwise find all six solutions to the equation $f(z) = 0$, giving answers in the form $z = re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$.

2

$$re^{3\theta i} = 8e^{2\pi/3 i} \text{ for } z^3 = -4 + 4\sqrt{3}i$$

$$r = 2 \quad 3\theta = 2\pi/3, 8\pi/3, -4\pi/3$$

$$\theta = 2\pi/9, 8\pi/9, -4\pi/9$$

$$re^{3\theta i} = 8e^{-2\pi/3 i} \text{ for } z^3 = -4 - 4\sqrt{3}i$$

$$r = 2 \quad 3\theta = -2\pi/3, -8\pi/3, 4\pi/3$$

$$\theta = -2\pi/9, -8\pi/9, 4\pi/9$$

$$\therefore z = 2e^{\pm 2\pi/9 i}, 2e^{\pm 4\pi/9 i}, 2e^{\pm 8\pi/9 i}$$

✓ any solution

✓ all solutions

(iii) Hence show that:

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{6\pi}{9} + \cos \frac{8\pi}{9} = -\frac{1}{2}$$

3

we know the roots of $z^6 + 8z^3 + 64 = 0$

are: $2e^{2\pi/9i}, 2e^{-2\pi/9i}, 2e^{4\pi/9i}, 2e^{-4\pi/9i}, 2e^{8\pi/9i}$ & $2e^{-8\pi/9i}$

$$\text{sum of roots} = -0/1$$

✓ sum of roots

$$\therefore 2[e^{2\pi/9i} + e^{-2\pi/9i} + e^{4\pi/9i} + e^{-4\pi/9i} + e^{8\pi/9i} + e^{-8\pi/9i}] = 0$$

$$\therefore 2[2\cos \frac{2\pi}{9} + 2\cos \frac{4\pi}{9} + 2\cos \frac{8\pi}{9}] = 0$$

$$\begin{aligned} \sqrt{e^{i\theta} + e^{-i\theta}} \\ = 2\cos \theta \end{aligned}$$

$$\therefore \cancel{4}[\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9}] = 0$$

$$\therefore \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{6\pi}{9} + \cos \frac{8\pi}{9} = \cos \frac{6\pi}{9}$$

added to
each side

$$\therefore \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{6\pi}{9} + \cos \frac{8\pi}{9} = -\frac{1}{2}$$

✓ answer /
exact value
of $\cos \frac{6\pi}{9}$

- (b) Mr Berry confiscates an *Efficiency Coaching* booklet from a student (who should not have been doing their tutoring homework in class) and drops it from the window of B14.

The booklet lands on the playground exactly 1.5 seconds later.

The booklet weighs 0.2 kg and experiences a force due to air resistance of magnitude $0.4v$ Newtons, where v is its velocity.

Letting gravity be 9.8 m/s^2 , find to two decimal places:

- (i) The speed of the booklet when it hits the ground

3

$$F = ma$$
$$0.2g = 0.2g - 0.4v$$

$$a = g - 2v$$

$$\frac{dv}{dt} = g - 2v$$

✓ equation

↑ $0.4v$
↓ $0.2g$ let down be positive

$$\frac{dt}{dv} = \frac{1}{g - 2v}$$

$$\int_0^t dt = -\frac{1}{2} \int_0^v \frac{-2}{g - 2v}$$

✓ integral

$$t - 0 = -\frac{1}{2} [\ln|g - 2v| - \ln|g|]$$

$$t = -\frac{1}{2} \ln \left| \frac{g - 2v}{g} \right|$$

$$-2t = \ln \left| \frac{g - 2v}{g} \right|$$

$$\frac{g - 2v}{g} = e^{-2t}$$

✓ answer

$$g - 2v = g e^{-2t}$$

$$2v = g - g e^{-2t}$$

$$v = \frac{g}{2} - \frac{g e^{-2t}}{2}$$

when $t = 1.5$

$$v = 4.66 \text{ m/s}$$

(2. d.p.)

(ii) The percentage of its terminal velocity it reached at the moment of impact

1

$$\text{terminal velocity} = 4.9 \text{ m/s}$$

$$\therefore \frac{4.656 \dots}{4.9} \times 100 \% = 95.02\% \quad (2 \text{ d.p.})$$



(iii) The height the booklet was dropped from

3

$$\frac{dx}{dt} = \frac{g}{2} - \frac{g e^{-2t}}{2}$$

$$\int_0^x dx = \frac{1}{2} \int_0^{1.5} (g - g e^{-2t}) dt$$

✓ set up
integral

$$x - 0 = \frac{1}{2} \left[gt + \frac{g}{2} e^{-2t} \right]_0^{1.5}$$

✓ primitive

$$x = \frac{1}{2} \left[9.8(1.5) + \frac{9.8}{2} e^{-2(1.5)} - \left(0 + \frac{9.8}{2}(1) \right) \right]$$

$$x = 5.02 \text{ m} \quad (2 \text{ d.p.})$$

✓ answer

(c) Given that $z \in \mathbb{C}$ such that $\text{Im}(z) \neq 0$ and $\frac{z}{1+z^2}$ is a real number. Show that $|z| = 1$

2

as $\frac{z}{1+z^2}$ is real the $\frac{1+z^2}{z}$ is also real

$$\frac{1+z^2}{z} = z + \frac{1}{z}$$

$$= x + iy + \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$$

$$= x + iy + \frac{x-iy}{x^2+y^2}$$

$$= x + \frac{x}{x^2+y^2} + \left(y - \frac{y}{x^2+y^2}\right)i$$

$$\therefore y - \frac{y}{x^2+y^2} = 0$$

$\checkmark$ using $\text{Im}\left(\frac{1}{z}\right) = 0$

$$y \left(1 - \frac{1}{x^2+y^2}\right) = 0$$

$$\text{as } y \neq 0$$

$$1 - \frac{1}{x^2+y^2} = 0$$

$$1 = \frac{1}{x^2+y^2}$$

$$x^2+y^2 = 1$$

$$\therefore |z| = 1$$

$\checkmark$ answer

(c) Given that $z \in \mathbb{C}$ such that $\operatorname{Im}(z) \neq 0$ and $\frac{z}{1+z^2}$ is a real number. Show that $|z| = 1$

2

Alternative solution

$\frac{z}{1+z^2}$ is real

$$\therefore \frac{z}{1+z^2} = \overline{\frac{z}{1+z^2}}$$

$$= \frac{\bar{z}}{\overline{1+z^2}}$$

$$\frac{z}{1+z^2} = \frac{\bar{z}}{1+\bar{z}^2}$$

✓ using
conjugates

$$\therefore z + \underbrace{z \bar{z} \bar{z}} = \bar{z} + \underbrace{\bar{z} z z}$$

$$z + |z|^2 \bar{z} = \bar{z} + |z|^2 z$$

$$z - \bar{z} + |z|^2 \bar{z} - |z|^2 z = 0$$

$$z - \bar{z} + |z|^2 (\bar{z} - z) = 0$$

$$z - \bar{z} - |z|^2 (z - \bar{z}) = 0$$

$$(z - \bar{z})(1 - |z|^2) = 0$$

$$\therefore |z|^2 = 1 \quad \text{as } z = \bar{z} \text{ since } \operatorname{Im}(z) \neq 0$$

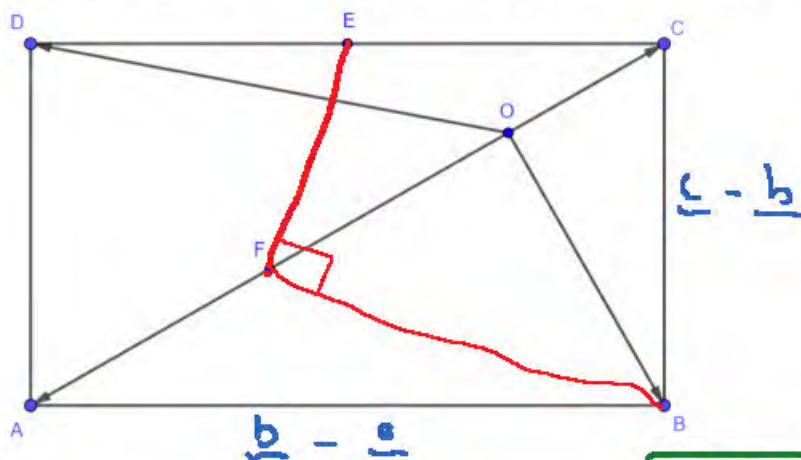
$$\therefore |z| = 1 \quad (\text{as } |z| > 0) \quad \boxed{\text{✓ answer}}$$

Question 15 (15 Marks)

- (a) In the diagram below, O is the origin and points A, B, C and D have position vectors $\underline{a}$, $\underline{b}$, $\underline{c}$ and $\underline{d}$ respectively.

E is the midpoint of CD and F is the midpoint of OA .

$ABCD$ is a rectangle such that O lies on diagonal AC , and OB is perpendicular to AC



- (i) Show that $|\underline{b}|^2 + \underline{a} \cdot \underline{c} = 0$

✓ dot product

2

$$\underline{AB} \cdot \underline{BC} = 0 \quad (\text{sides of rectangle perpendicular})$$

$$(\underline{b} - \underline{a}) \cdot (\underline{c} - \underline{b}) = 0$$

$$\cancel{\underline{b} \cdot \underline{c}} - \underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{c} + \cancel{\underline{a} \cdot \underline{b}} = 0$$

$$\therefore -|\underline{b}|^2 - \underline{a} \cdot \underline{c} = 0 \quad (\text{as } \underline{a} \perp \underline{b} \text{ \& } \underline{b} \perp \underline{c})$$

$$(\times -1) \therefore |\underline{b}|^2 + \underline{a} \cdot \underline{c} = 0$$

✓ answer

- (ii) Hence, or otherwise, show that $\angle BFE = \frac{\pi}{2}$

3

$$\underline{BF} = \underline{BO} + \underline{OF}$$

$$= -\underline{b} + \frac{1}{2}\underline{a}$$

$$= \frac{1}{2}\underline{a} - \underline{b}$$

✓ $\underline{BF}, \underline{FE}$

$$\underline{FE} = \underline{FO} + \underline{OC} + \underline{CE}$$

$$= -\frac{1}{2}\underline{a} + \underline{c} + \frac{1}{2}(\underline{a} - \underline{b})$$

$$\underline{OC} = \underline{AO}$$

$$= -\cancel{\frac{1}{2}\underline{a}} + \underline{c} + \cancel{\frac{1}{2}\underline{a}} - \frac{1}{2}\underline{b}$$

$$= \underline{c} - \frac{1}{2}\underline{b}$$

$$\therefore \underline{BF} \cdot \underline{FE} = \left(\frac{1}{2}\underline{a} - \underline{b}\right) \cdot \left(\underline{c} - \frac{1}{2}\underline{b}\right)$$

✓ dot product

$$= \frac{1}{2}\underline{a} \cdot \underline{c} - \frac{1}{4}\underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{c} + \frac{1}{2}\underline{b} \cdot \underline{b}$$

✓ result from (i)

$$= \frac{1}{2}(\underline{c} \cdot \underline{c} + |\underline{b}|^2)$$

$$= 0 \quad \therefore \underline{BF} \perp \underline{FE}$$

(b) (i) Expand and simplify $3[(x-2)^3 + 2]$

1

$$\begin{aligned} & 3 [x^3 - 3(2)x^2 + 3(2)^2x - \overbrace{2^3}^6 + 2] \\ &= 3 [x^3 - 6x^2 + 12x - 6] \\ &= 3x^3 - 18x^2 + 36x - 18 \end{aligned}$$



(ii) Given

$$I_n = \int_0^a \frac{x^n}{\sqrt{a^2 - x^2}} dx, \quad n \in \mathbb{N}$$

clearly show that:

$$I_n = \frac{a^2(n-1)}{n} I_{n-2}, \quad n \geq 2$$

3

$$\begin{aligned} I_n &= \int_0^a \underbrace{x^{n-1}}_v \times \underbrace{x(a^2 - x^2)^{-\frac{1}{2}}}_{u'} dx \\ &= \left[\underbrace{x^{n-1}}_v \times \underbrace{-(a^2 - x^2)^{\frac{1}{2}}}_u \right]_0^a - \int_0^a \underbrace{(n-1)}_{v'} \underbrace{x^{n-2}}_v \times \underbrace{-(a^2 - x^2)^{-\frac{1}{2}}}_u dx \\ &= (n-1) \int_0^a x^{n-2} \sqrt{a^2 - x^2} dx \times \frac{\sqrt{a^2 - x^2}}{\sqrt{a^2 - x^2}} \\ &= (n-1) \int_0^a \frac{x^{n-2} (a^2 - x^2)}{\sqrt{a^2 - x^2}} dx \\ &= (n-1) \left[\int_0^a \frac{a^2 x^{n-2}}{\sqrt{a^2 - x^2}} dx - \int_0^a \frac{x^n}{\sqrt{a^2 - x^2}} dx \right] \end{aligned}$$

✓ integration by parts

$$I_n = (n-1) [a^2 I_{n-2} - I_n]$$

✓ I_n & I_{n-2}

$$\cancel{I_n} = a^2(n-1) I_{n-2} + \cancel{n I_n} - a^2 I_{n-2} + \cancel{I_n}$$

$$n I_n = a^2 I_{n-2} (n-1)$$

$$\therefore I_n = \frac{a^2(n-1)}{n} I_{n-2}$$

✓ answer

(iii) Show that

$$\int_2^4 \frac{3x^3 - 18x^2 + 36x - 18}{\sqrt{4x - x^2}} dx = 3I_3 + 3\pi$$

$$\text{LHS} = \int_2^4 \frac{3[(x-2)^3 + 2]}{\sqrt{4x - x^2}} dx \quad (\text{from (i)})$$

✓ substitution

$$= 3 \int_2^4 \frac{(x-2)^3 + 2}{\sqrt{4 - (4 - 4x + x^2)}} dx$$

$$= 3 \int_2^4 \frac{(x-2)^3 + 2}{\sqrt{4 - (x-2)^2}} dx$$

$$u = x - 2$$

$$du = dx$$

$$\text{when } x = 4$$

$$u = 2$$

$$\text{when } x = 2$$

$$u = 0$$

$$= 3 \int_0^2 \frac{u^3 + 2}{\sqrt{2^2 - u^2}} du$$

$$= 3 \int_0^2 \frac{u^3}{\sqrt{2^2 - u^2}} du + 3 \int_0^2 \frac{2}{\sqrt{2^2 - u^2}} du$$

✓ complete square on denominator

$$= 3I_3 + 6 \int_0^2 \frac{1}{\sqrt{2^2 - u^2}} \quad (\text{where } a=2)$$

$$= 3I_3 + 6 \left[\sin^{-1}\left(\frac{u}{2}\right) \right]_0^2$$

✓ use recurrence relation

$$= 3I_3 + \left[\sin^{-1}\left(\frac{2}{2}\right) - \sin^{-1}\left(\frac{0}{2}\right) \right]$$

$$= 3I_3 + 6 \left(\frac{\pi}{2} \right)$$

✓ integrate to get $\sin^{-1} \frac{u}{2}$

$$= 3I_3 + 3\pi$$

$$= \text{RHS}$$

(iv) Hence or otherwise, evaluate

$$\int_2^4 \frac{3x^3 - 18x^2 + 36x - 18}{\sqrt{4x - x^2}} dx$$

2

$$3I_3 + 3\pi$$

$$= \cancel{3} \left(\frac{2^2(3-1)}{\cancel{3}} \right) I_1 + 3\pi$$

$$= 8I_1 + 3\pi$$

$$= 8 \int_0^2 \frac{x}{\sqrt{4-x^2}} dx + 3\pi$$

$$= 8 \left[-\sqrt{4-x^2} \right]_0^2 + 3\pi$$

$$= 8 \left[-\cancel{\sqrt{4-4}} - (-\sqrt{4-0^2}) \right] + 3\pi$$

$$= 8(2) + 3\pi$$

$$= 16 + 3\pi$$

✓ using
reduction
formula

✓ correct
answer

Question 16 (15 Marks)

- (a) (i) By considering the expansion of $k^3 - (k-1)^3$ show that:

$$\sum_{k=1}^n k^2 = \frac{n}{6}(n+1)(2n+1)$$

2

(do **NOT** use mathematical induction)

$$k^3 - (k-1)^3 = \cancel{k^3} - (\cancel{k^3} - 3k^2 + 3k - 1) \\ = 3k^2 - 3k + 1$$

$$\therefore 3k^2 = k^3 - (k-1)^3 + 3k - 1$$

$$\therefore 3 \sum_{k=1}^n k^2 = \sum_{k=1}^n \underbrace{k^3 - (k-1)^3}_{\text{telescoping}} + \sum_{k=1}^n \underbrace{3k - 1}_{\text{AP}}$$

$$3 \sum_{k=1}^n k^2 = \cancel{1^3} - 0^3 + \cancel{2^3} - \cancel{1^3} + \cancel{3^3} - \cancel{2^3} + \dots + n^3 + \cancel{(n-1)^3} \\ + \frac{n}{2} [\underbrace{2(2)}_{2a} + \underbrace{(n-1)3}_d]$$

$$= n^3 + \frac{n}{2} [4 + 3n - 3]$$

$$= n^3 + \frac{n}{2} [3n + 1]$$

$$= \frac{n}{2} [2n^2 + 3n + 1]$$

$$= \frac{n}{2} [2n^2 + 2n + n + 1]$$

$$= \frac{n}{2} [2n(n+1) + 1(n+1)]$$

$$3 \sum_{k=1}^n k^2 = \frac{n}{2} (n+1)(2n+1)$$

$$\sum_{k=1}^n k^2 = \frac{n}{6} (n+1)(2n+1)$$

✓ telescoping series

✓ AP

Assume that such a set exists, where the primes are numbered:

$$p_1, p_2, p_3, \dots, p_n$$

$$\text{now, } 2^2 + 3^2 + \dots + n^2 = \frac{n}{6} (n+1)(2n+1) - 1$$

(by rearranging part (i))

$$\text{let } S = p_1 p_2 \dots p_n$$

$\& \quad n = 6S$

✓ substitution

$$\therefore 2^2 + 3^2 + \dots + n^2 = \frac{6S}{6} (S+1)(2S+1) - 1$$
$$= S(S+1)(2S+1) - 1$$

as $S(S+1)(2S+1)$ is a multiple of all of the elements in the set

$S(S+1)(2S+1) - 1$ is not a multiple of any element $\&$ there is always S on n such that $2^2 + 3^2 + \dots + n^2$ is not a multiple of any element of the set

✓ explanation

(b) (i) Let $\underline{w} = \underline{u} + r\underline{v}$ where $\underline{w}$, $\underline{u}$ and $\underline{v}$ are vectors and r is a scalar.

Prove that $|\underline{v}|^2 r^2 + 2(\underline{u} \cdot \underline{v})r + |\underline{u}|^2 \geq 0$

1

$$\begin{aligned}
 & |\underline{v}|^2 r^2 + 2(\underline{u} \cdot \underline{v})r + |\underline{u}|^2 \\
 &= \underline{v} \cdot \underline{v} r^2 + 2(\underline{u} \cdot \underline{v})r + \underline{u} \cdot \underline{u} \\
 &= (\underline{v}r + \underline{u}) \cdot (\underline{v}r + \underline{u}) \\
 &= \underline{w} \cdot \underline{w} \\
 &= |\underline{w}|^2 \\
 &\geq 0
 \end{aligned}$$



(ii) Use your result from (i) to show that $\underline{u} \cdot \underline{v} \leq |\underline{u}||\underline{v}|$

2

$$\text{as } \underbrace{|\underline{v}|^2}_a r^2 + \underbrace{2(\underline{u} \cdot \underline{v})}_b r + \underbrace{|\underline{u}|^2}_c \geq 0$$

$$\Delta \leq 0$$

$$\therefore [2(\underline{u} \cdot \underline{v})]^2 - 4|\underline{v}|^2 |\underline{u}|^2 \leq 0$$

$$\therefore \cancel{4}(\underline{u} \cdot \underline{v})^2 \leq \cancel{4}|\underline{u}|^2 |\underline{v}|^2$$

$$(\underline{u} \cdot \underline{v})^2 \leq (|\underline{u}||\underline{v}|)^2$$

$$(\text{as } |\underline{u}||\underline{v}| \geq 0)$$

$$\therefore \underline{u} \cdot \underline{v} \leq |\underline{u}||\underline{v}|$$

✓ discriminant

✓ answer

(iii) Use your result from part (ii) to show that for real numbers x, y and z .

$$\text{let } xy + yz + zx \leq x^2 + y^2 + z^2$$

1

$$\underline{u} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$xy + yz + zx \leq \left| \sqrt{x^2 + y^2 + z^2} \right| \left| \sqrt{x^2 + y^2 + z^2} \right|$$

$$\leq x^2 + y^2 + z^2$$

(from ii)



$$\underline{v} = \begin{pmatrix} y \\ z \\ x \end{pmatrix}$$

(iv) Use your result from part (iii) to show that for non-negative values of x, y and z :

$$xyz \leq \frac{x^3 + y^3 + z^3}{3}$$

1

$$(xy + yz + zx)(x + y + z) \leq (x^2 + y^2 + z^2)(x + y + z)$$

$$x(x + y + z) \quad x(x + y + z)$$

$$\therefore \cancel{x^2y} + \cancel{xy^2} + \cancel{z^2x} + \cancel{zx^2} + \cancel{y^2z} + \cancel{zy^2} + \cancel{z^2y} + \cancel{yz^2}$$

$$\leq x^3 + \cancel{x^2y} + \cancel{x^2z} + \cancel{xy^2} + y^3 + \cancel{y^2z} + \cancel{yz^2} + \cancel{z^2x} + \cancel{zx^2} + z^3$$

$$\therefore 3xyz \leq x^3 + y^3 + z^3$$

$$\therefore xyz \leq \frac{x^3 + y^3 + z^3}{3}$$



(v) Given that $a + b + c + d = 0$ and $a^2 + b^2 + c^2 + d^2 = 12$. Find:

2

1. The maximum value of $abcd$

as max value of $abcd > 0$
we can find max value
of $|a||b||c||d|$ which is the same
as the max value of $|a||b||c||d|$

$$\sqrt{|a||b||c||d|} = \frac{|a||b|}{2} + \frac{|c||d|}{2}$$

$$= \frac{\frac{a^2 + b^2}{2}}{2} + \frac{\frac{c^2 + d^2}{2}}{2}$$

$$= \frac{a^2 + b^2 + c^2 + d^2}{4}$$

$$= \frac{12}{4}$$

$$= 3$$

$$\therefore \text{max value of } abcd = 9$$

✓ 2 applications
of AM/GM
inequality

3²



✓ answer

$$a+b+c+d = 0 \quad (\text{given})$$

$$\therefore d = -(a+b+c)$$

$$\therefore abcd = -abc(a+b+c)$$

✓ rearrange
to find
max

minimising $-abc(a+b+c)$ is the same as maximising $abc(a+b+c)$.

$$a^2+b^2+c^2+d^2 = 12 \quad (\text{given})$$

$$\therefore a^2+b^2+c^2 + [- (a+b+c)]^2 = 12 \quad (\text{sub in } d)$$

$$\therefore a^2+b^2+c^2 + \underline{a^2+b^2+c^2 + 2ab+2bc+2ca} = 12$$

$$\therefore 2(a^2+b^2+c^2 + ab+bc+ca) = 12$$

$$a^2+b^2+c^2 + ab+bc+ca = 6 \quad (1)$$

$$\begin{aligned} (a+b+c)^2 &= a^2+b^2+c^2 + 2ab+2bc+2ca \\ &= a^2+b^2+c^2 + ab+bc+ca + ab+bc+ca \\ &= 6 + ab+bc+ca \quad (\text{from } (1)) \end{aligned}$$

$$= 6 + \frac{ab+bc+ca + ab+bc+ca}{2}$$

✓ use
(iii)

$$\leq 6 + \frac{a^2+b^2+c^2 + ab+bc+ca}{2} \quad (\text{from } (iii))$$

$$= 6 + \frac{6}{2} \quad (\text{from } (1))$$

$$= 9$$

$$\therefore (a+b+c)^2 \leq 9 \quad (2)$$

$$\sqrt[3]{abc} \leq \frac{a+b+c}{3} \quad \left(\text{from (iv) where } x = \sqrt[3]{a}, y = \sqrt[3]{b}, z = \sqrt[3]{c} \right)$$

$$\therefore abc \leq \frac{(a+b+c)^3}{27}$$

✓ use (iv)

$$abc(a+b+c) \leq \frac{(a+b+c)^4}{27}$$

$\times (a+b+c)$ $\times (a+b+c)$

$$= \frac{[(a+b+c)^2]^2}{27}$$

$$\leq \frac{(9)^2}{27}$$

$$= \frac{81}{27}$$

$$= 3$$

note that a minimum value occurs with 3 positive & 1 negative so we take the case where $a, b, c > 0$

as the maximum value of $abc(a+b+c)$ is 3 the minimum value of $-abc(a+b+c)$ must be -3

$\therefore$ minimum value of $abcd$ is -3

✓ answer